6. Data Design

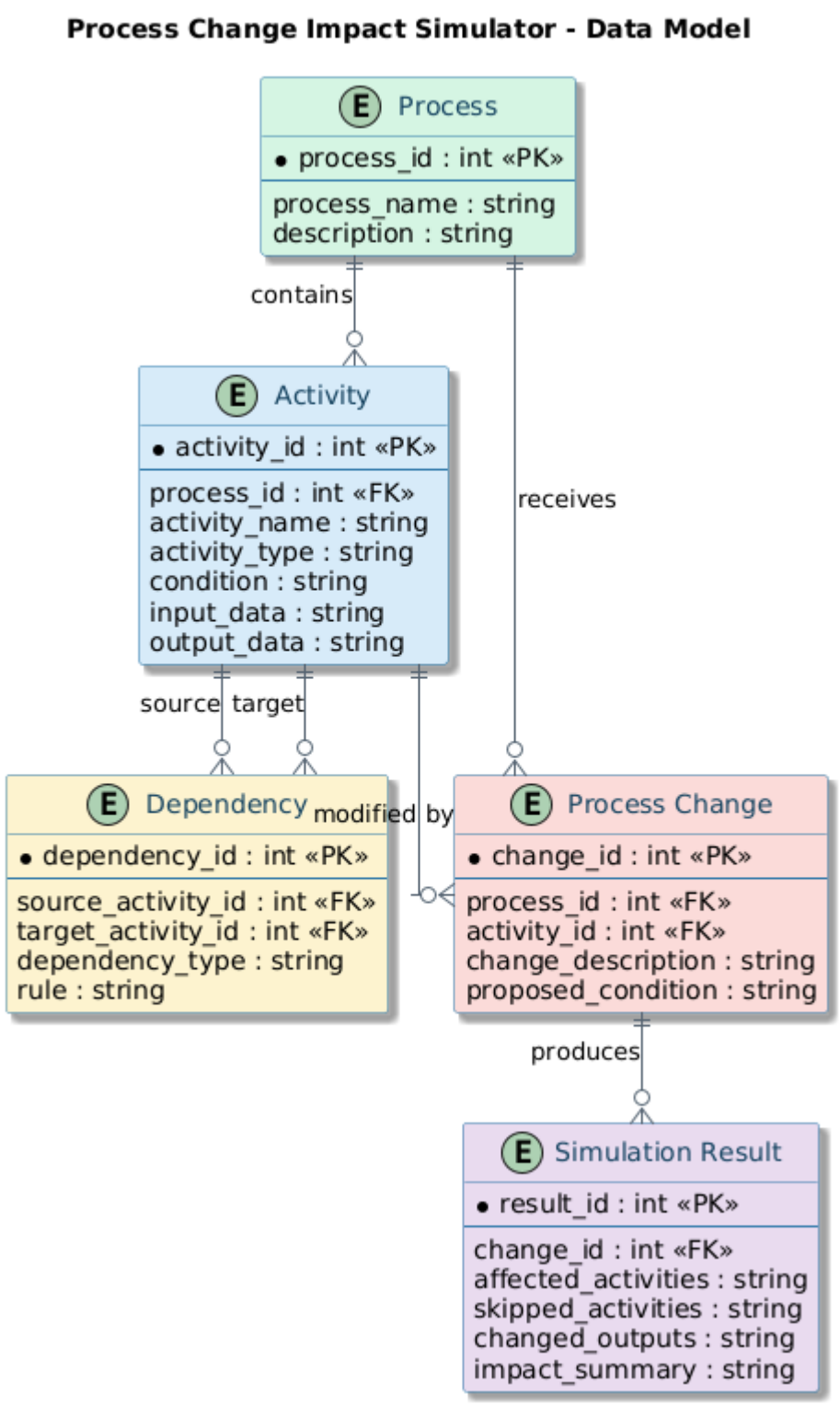
The Process Change Impact Simulator will use SQLite to store process information, activity details, dependencies, proposed changes, and simulation results. Each table will have a primary key to identify its records. Foreign keys will connect related records and help maintain data consistency.

Main database tables

Table	Important attributes	Purpose
Process	`process_id` (PK), `process_name`, `description`	Stores business process details.
Activity	`activity_id` (PK), `process_id` (FK), `activity_name`, `activity_type`, `condition`, `input_data`, `output_data`	Stores activities and their conditions, inputs, and outputs.
Dependency	`dependency_id` (PK), `source_activity_id` (FK), `target_activity_id` (FK), `dependency_type`, `rule`	Defines relationships between activities.
Process Change	`change_id` (PK), `process_id` (FK), `activity_id` (FK), `change_description`, `proposed_condition`	Records proposed changes to a process.
Simulation Result	`result_id` (PK), `change_id` (FK), `affected_activities`, `skipped_activities`, `changed_outputs`, `impact_summary`	Stores the results of comparing original and modified workflows.

Relationships and data management

- One process can contain multiple activities and dependencies.
- One process can have multiple proposed changes.
- Each dependency connects a source activity to a target activity.
- Each proposed change can produce simulation results.
- Foreign keys will help prevent records from referring to missing processes or activities. Required fields and validation rules will help prevent incomplete records.
- Process definitions can be imported and exported as JSON files.


Process Change Impact Simulator - Data Model
Process
process_id : int «PK»
process_name : string
description : string
contains
receives
Activity
activity_id : int «PK»
process_id : int «FK»
activity_name : string
activity_type : string
condition : string
input_data : string
output_data : string
source
target
Dependency
dependency_id : int «PK»
source_activity_id : int «FK»
target_activity_id : int «FK»
dependency_type : string
rule : string
modified by
Process Change
change_id : int «PK»
process_id : int «FK»
activity_id : int «FK»
change_description : string
proposed_condition : string
produces
Simulation Result
result_id : int «PK»
change_id : int «FK»
affected_activities : string
skipped_activities : string
changed_outputs : string
impact_summary : string

ER Diagram